Human Activity Recognition: A Comparative Analysis of Machine Learning Models

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Abstract

Smart devices, such as smart watches, rings, wristbands, and other accessories, have become increasingly popular over the last few years. As these devices have become more accurate and have expanded their capabilities, many rely on them for important health data. Given the necessity for health data to be highly accurate, this project sets out to understand the accuracy with which different machine learning models can identify different movements and activities. The dataset used is the UCI Human Activity Recognition dataset, which contains 561 features derived from gyroscope and accelerometer readings. It contains 6 different activities: sitting, standing, walking, laying, walking upstairs, and walking downstairs. Our results show that all of the models were able to differentiate between the different activities with high accuracy, with the histogram gradient boosting model performing best across all metrics. We were able to achieve and maintain high accuracy after significant feature reduction, which showed strong improvement in terms of efficiency and execution time.

Keywords

Machine Learning, Human Activity Recognition, Sensor Data, Feature Selection, Hyperparameter Tuning

Contents

1 Preliminary Analysis

1.1 Goals

The goal of this project is to compare the performance of multiple machine learning algorithms on classifying six distinct activities: walking, walking upstairs, walking downstairs, sitting, standing, and Laying.

As wearable technology (smartwatches, fitness trackers, smart rings, etc.) becomes ubiquitous, the ability to accurately categorize user behavior in real-time has immense applications in healthcare, elderly monitoring, and fitness tracking. For this data to be safe and effective, it must be highly accurate. Especially in a time where smart devices can be used to make medical diagnoses or alert people of possible health issues, the accuracy of these results is vital. We set out to understand how effective these models are at understanding health sensor data, and which techniques should be used moving forward in this space.

1.2 Data

The data was sourced from the UCI Machine Learning Repository (Human Activity Recognition Using Smartphones). It is comprised of 7,352 entries.

Each row represents a window of sensor data captured at 50Hz. The dataset contains 561 features derived from the raw time-domain signals (prefixed with 't') and frequency-domain signals (prefixed with 'f'). These features include statistical measures such as mean, standard deviation, signal magnitude area (SMA), energy, and entropy for various axes (X, Y, Z).

The data also already included some preprocessing, which consisted of a median filter and a 3rd order low pass Butterworth filter with a corner frequency of 20 Hz to remove noise. Additionally, the acceleration signal was separated into the body acceleration and gravity acceleration signals using another low pass Butterworth filter with a corner frequency of 0.3 Hz.

1.3 Preprocessing

1.3.1 Label Mapping

The original dataset encoded activities as integers 1-6. To make the analysis interpretable, we mapped these integers to their semantic labels: 1: WALKING, 2: WALKING_UPSTAIRS, 3: WALKING_DOWNSTAIRS, 4: SITTING, 5: STANDING, 6: LAYING. As required by some of the models, we converted the labels to range from 0-5 instead of 1-6, and did so for all of the models to ensure consistency.

1.3.2 Scaling

Since the dataset includes features with varying magnitudes (e.g., energy vs. coefficients), we applied `StandardScaler` to normalize the data. This transformation ensures that models sensitive to magnitude, such as SVM and Neural Networks, effectively converge.

1.4 Data Visualization and Exploratory Analysis

As part of our exploratory analysis, we wanted to understand how the different activities might be separable for different features. The following plots were important for understanding how the data might be separable and which activities might be the most difficult to differentiate

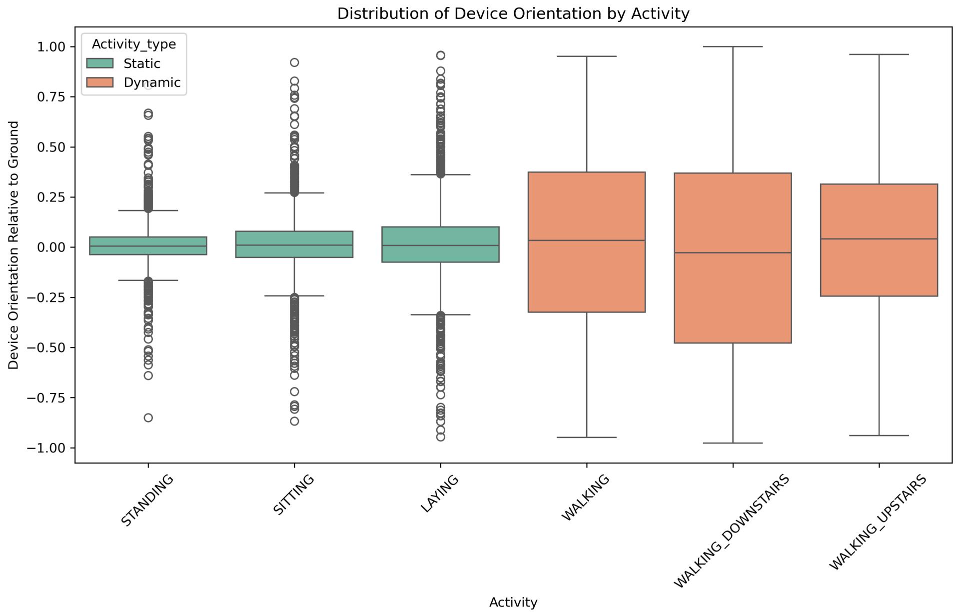

Figure 1: Boxplot of Device Orientation Relative to Ground by Activity and Activity Type

Looking at the orientation of the device relative to the ground (calculated by the angle between the vector of the average acceleration on the x-axis and the gravity vector), it is apparent that the variance for dynamic activities (i.e. walking, walking upstairs, and walking downstairs) is much higher than that of the static activities. The 25th and 75th percentiles for the static activities are very tight around the median, while those of the dynamic activities are significantly farther. While this is only one of many features, it demonstrates how differentiating between activity type might be trivial, but differentiating within type or between specific movements could be challenging

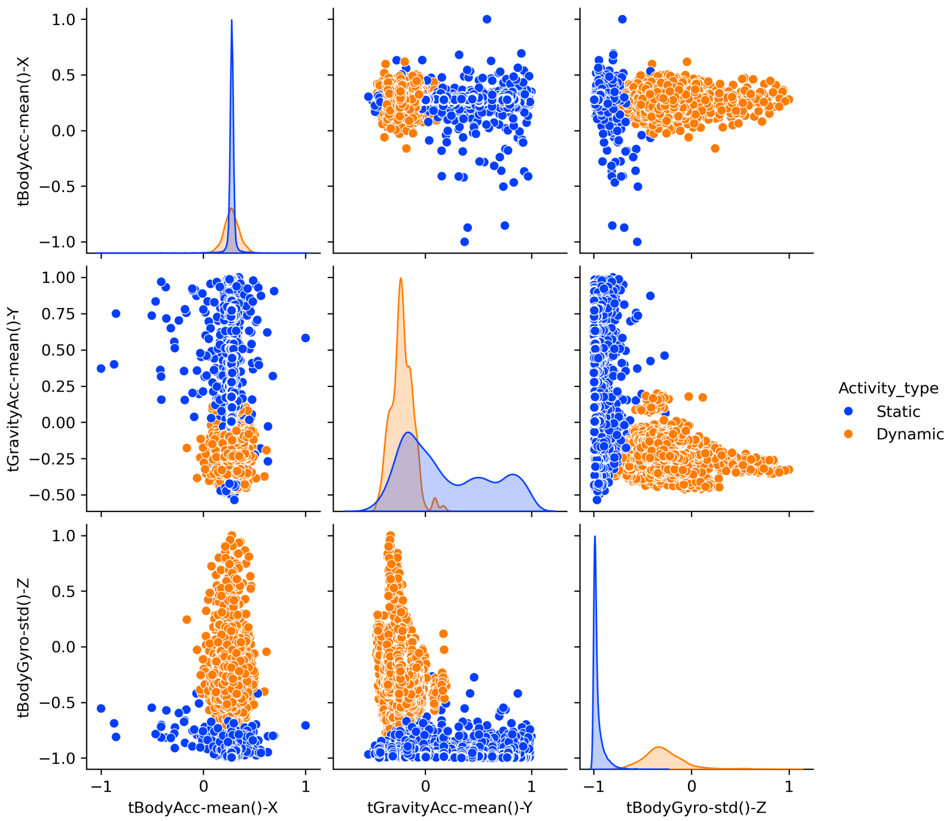

Figure 2: Relationships Between Different Variables Colored by Activity Type

Diving further into the separation between static and dynamic activities, we see that this difference holds between a number of different randomly selected variables. While there is overlap among some, we see clear separation between the two activity types, both in terms of value and variance (for the plots between two of the same variable). Based on these plots, we expect our models to be very successful in classifying the correct activity type.

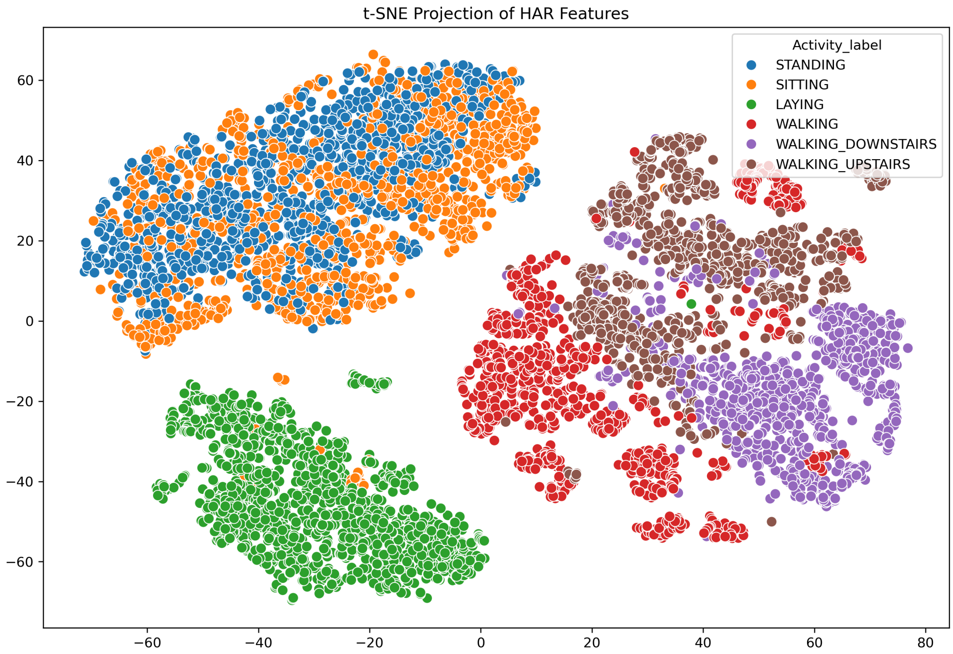

Figure 3: t-SNE Projection of the Dataset, Colored by Activity

Finally, we wanted to understand how separable the different activities were after combining all of the data. To do this, we used a t-distributed stochastic neighbor embedding (t-SNE) projection of the data to visualize the groupings of different activities on a 2-dimensional plane. Since t-SNE uses all of the features and reduces them down to 2-dimensions, it is an informative way to see how the different activities can be separated using the data. What we observe is a clear formation of 3 groups: sitting and standing, laying, and walking, walking upstairs, and walking downstairs. This makes intuitive sense, as one would expect the walking activities to all be similar, sitting and standing to be very similar, and laying to be relatively unique. Within the groupings, sitting and standing seems to be most similar, while there appears to be some level of separation between the different walking activities. Overall, this visualization was encouraging in that it showed that the activities were for the most part, separable.

2 Methodology

2.1 Feature Selection

The analysis was done in two parts: first, with some basic domain-knowledge driven filtering of features which only kept features of certain types, such as aggregated features. Then, a more robust feature selection was done, which involved correlation filtering and feature importance.

2.1.1 First Analysis

1. **Domain-informed Feature Selection:** We decided to reduce the number of features in the data set at first by removing redundancy in the data. For example, aggregate features (which ended in -mean() or -std()) would contain all of the information from the raw data. Additionally, we kept features that contained Mag- (for

magnitude) and angle() (an angle between two vectors), as these would be essential for dividing between certain activities, such as static activities where movement is not a factor. After this step, we were left with 173 features from the original 561.

2.1.2 Second Analysis

1. **Correlation Filter:** We calculated the correlation matrix for the training set and removed features with a Pearson correlation coefficient greater than 0.95. This removed redundant signals (e.g., body acceleration mean X and gravity mean X often move together). This step removed 101 correlated features, leaving a total of 71.

2. **Feature Importance:** We trained a Random Forest Classifier on the remaining features to calculate the "Gini Importance." We selected the top features that contributed most to the predictive power of the model. This step reduced the feature space to only 23 from the 71 selected in the prior step. We felt that this set was sufficiently small to proceed to the second analysis without doing any further feature reduction. The results of the feature importance evaluation are below:

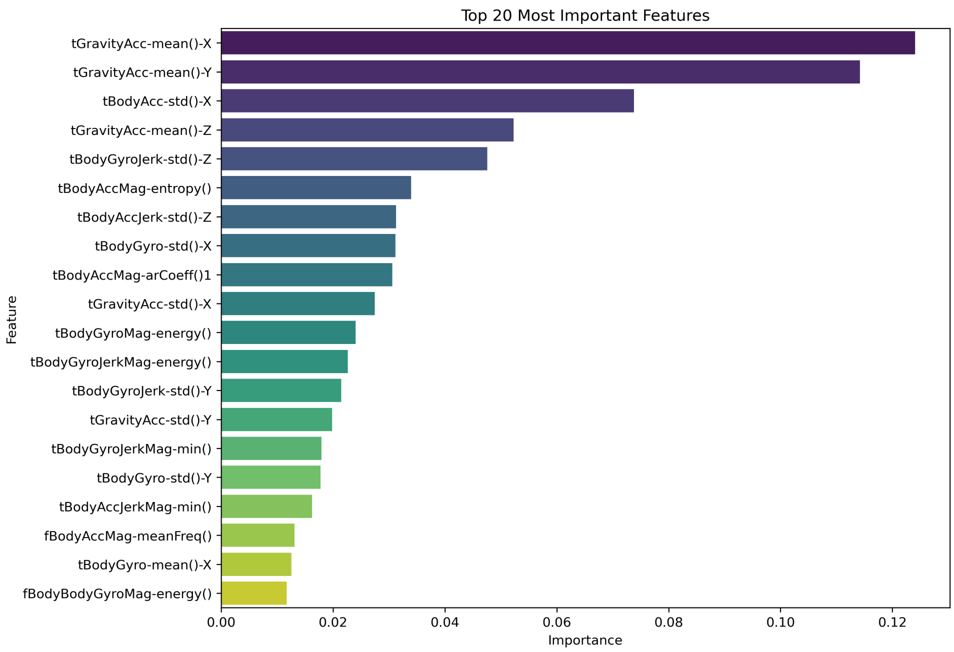

Figure 4: 20 Most Important Features Based on Random Forest Gini Importance

2.2 Hyperparameter Tuning

In the second analysis, we performed hyperparameter tuning for the Multi-Layer Perceptron (MLP) model to evaluate which combination of hidden layers, hidden layer sizes, learning rate, and dropout rate was optimal. In the end, this made very little difference in the performance of the model on the data, and actually landed on the default parameters used for the model. This suggests that any lack in performance is likely due to the MLP model itself, and not due to inefficient hyperparameters.

3 Modeling

3.1 Models

We evaluated six distinct algorithms to cover linear, tree-based, and deep learning approaches:

3.1.1 Logistic Regression

Used as a baseline to test the linear separability of the dynamic activities.

3.1.2 Random Forest Classification

An ensemble method chosen for its robustness to noise and ability to handle the large feature space.

3.1.3 Support Vector Machine (SVM)

Implemented with an RBF kernel. SVMs are traditionally state-of-the-art for HAR tasks due to their ability to find complex hyperplanes in high-dimensional space.

3.1.4 Gradient Boosting Classifiers

We evaluated two variations of gradient boosting to analyze the trade-off between training speed and accuracy:

1. Standard Gradient Boosting: We implemented the standard `GradientBoostingClassifier`, which builds an additive model in a forward stage-wise fashion. It optimizes a loss function by sequentially adding weak learners (decision trees), where each new tree tries to correct the residual errors of the previous ones. However, this method requires sorting all feature values for every potential split, which proved computationally expensive given the high dimensionality.

2. Histogram-Based Gradient Boosting: To address the computational costs, we implemented the `HistGradientBoostingClassifier`. This algorithm (inspired by LightGBM) discretizes continuous feature values into integer-valued bins (histograms). By splitting on these bins rather than sorting every single data point, the algorithm reduces the complexity of finding splits, resulting in significantly faster training times while maintaining comparable accuracy.

3.1.5 Neural Network (MLP)

A Multi-Layer Perceptron was implemented using Keras/TensorFlow. The architecture consisted of two hidden layers (64 and 32 neurons) with ReLU activation and Dropout (0.2) to prevent overfitting.

4 Results

All of the models achieved high accuracy ($> 94\%$) in classifying the correct activity. However, there was some variance in performance and execution time between the models.

4.1 First Analysis

Using the 173 features from the initial feature screening, we achieved the following results:

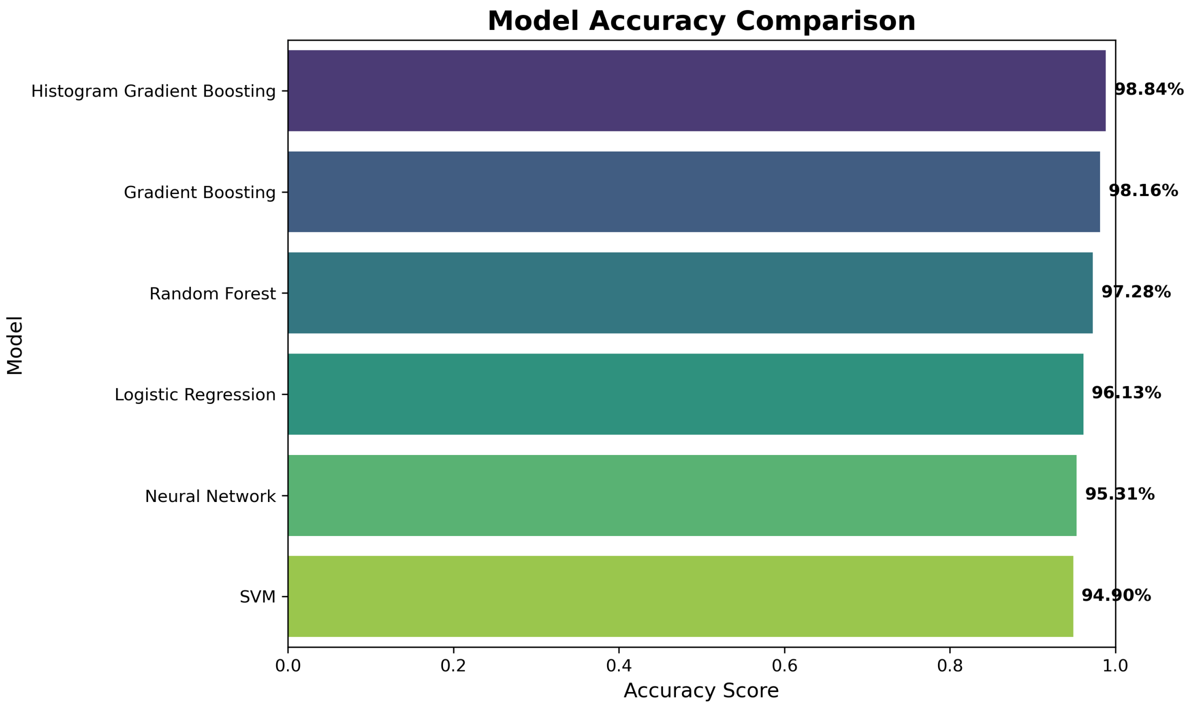

Figure 5: Model Accuracy

While all models had strong performance, the gradient boosting algorithms performed the best, with both achieving over 98% accuracy. The SVM proved to be the worst relative to the others, with a 95% accuracy.

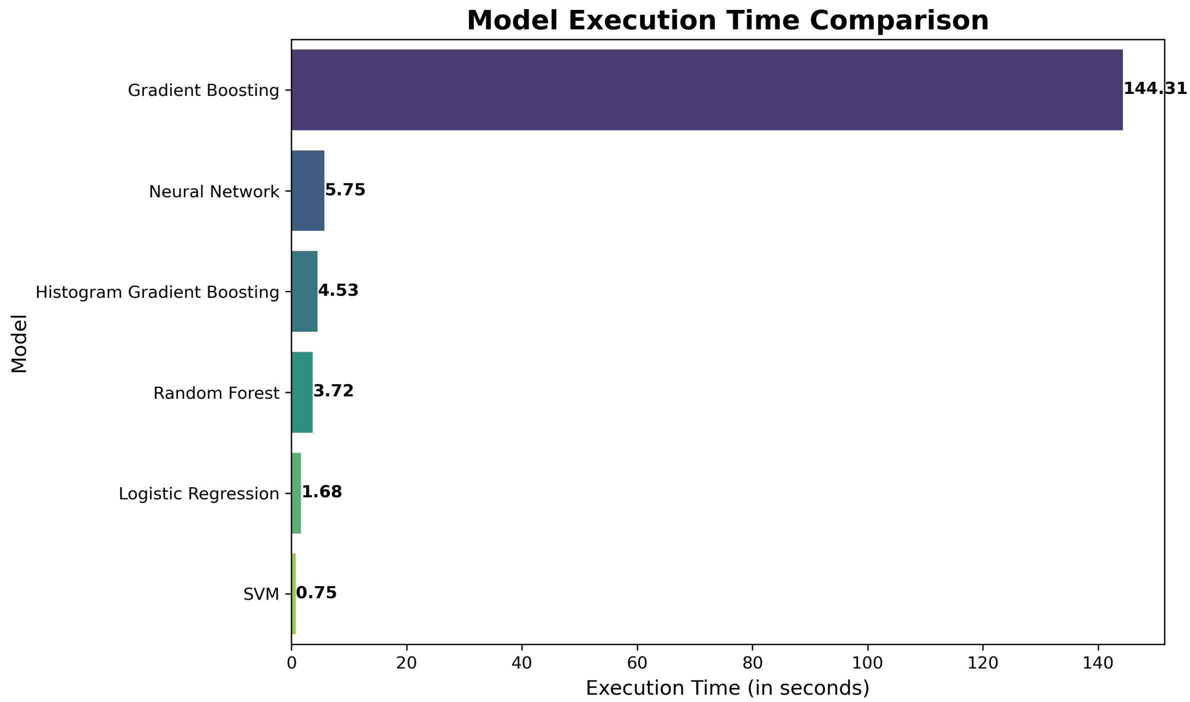

Figure 6: Execution Time

Looking at the execution times, the standard gradient boosting algorithm stands out as the most inefficient and expensive from an execution-time perspective. Moreover, it shows the value of the Histogram Gradient Boosting model in optimizing the process massively while maintaining (and in this case, outperforming) the gradient boosting model. The rest of the models had relatively low execution times, especially considering the high dimensional feature space.

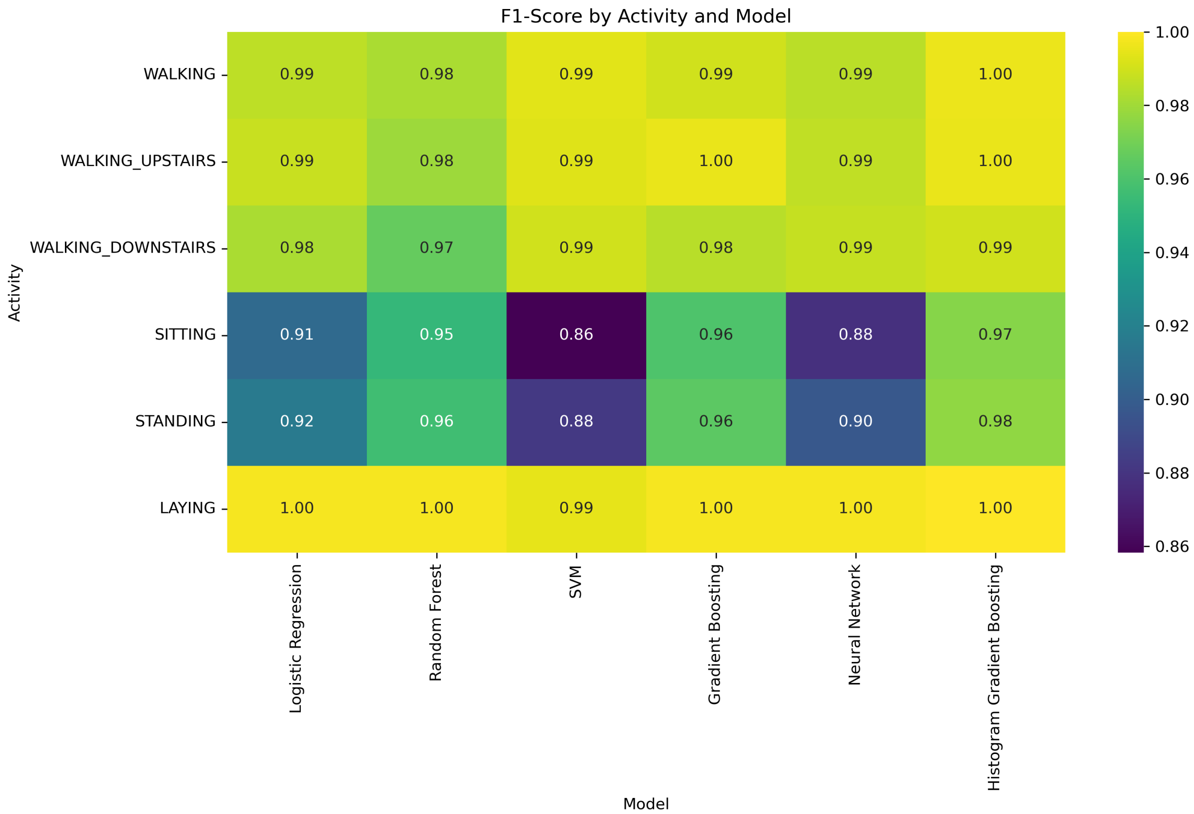

Figure 7: F1 Scores

The F1 scores allow us to better understand how each model is performing at classifying each activity type. As we can see, differentiating between sitting and standing is by far the most difficult task for the models, but 3 of them perform exceptionally well: random forest, gradient boosting, and histogram gradient boosting. The ensemble models perform significantly better than the others, and this is likely due to their ability to draw complex, non-linear decision boundaries around the categories. For all other activities, the performance is strong across the board, with near 100% accuracy for every model.

4.2 Second Analysis

Given the strong performance in the first comparison, we wanted to see how greatly reducing the feature space would affect performance, and whether some models would be more strongly affected than others.

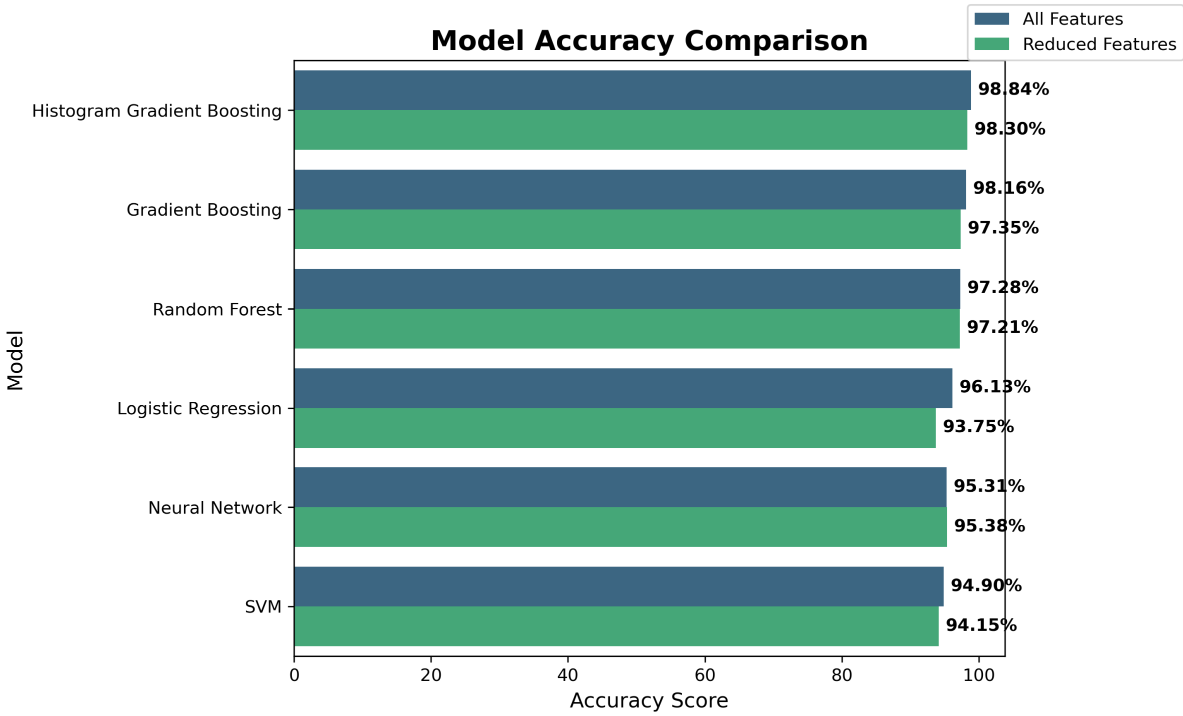

Figure 8: Model Accuracy

We see that, despite the greatly reduced feature space (an 87% reduction), all models maintain their strong performance. While almost all experience a slight reduction in performance (the outlier being the MLP model, which performs slightly better and is likely due to the slight improvement from the hyperparameter tuning used), the reduction is very small in nearly every case. The best model, the histogram gradient boosting model, only loses about 0.05% accuracy after the feature reduction. The logistic regression model experiences the greatest drop in accuracy (about 2.5%), which is likely due to its inability to learn more complex boundaries between the smaller feature set.

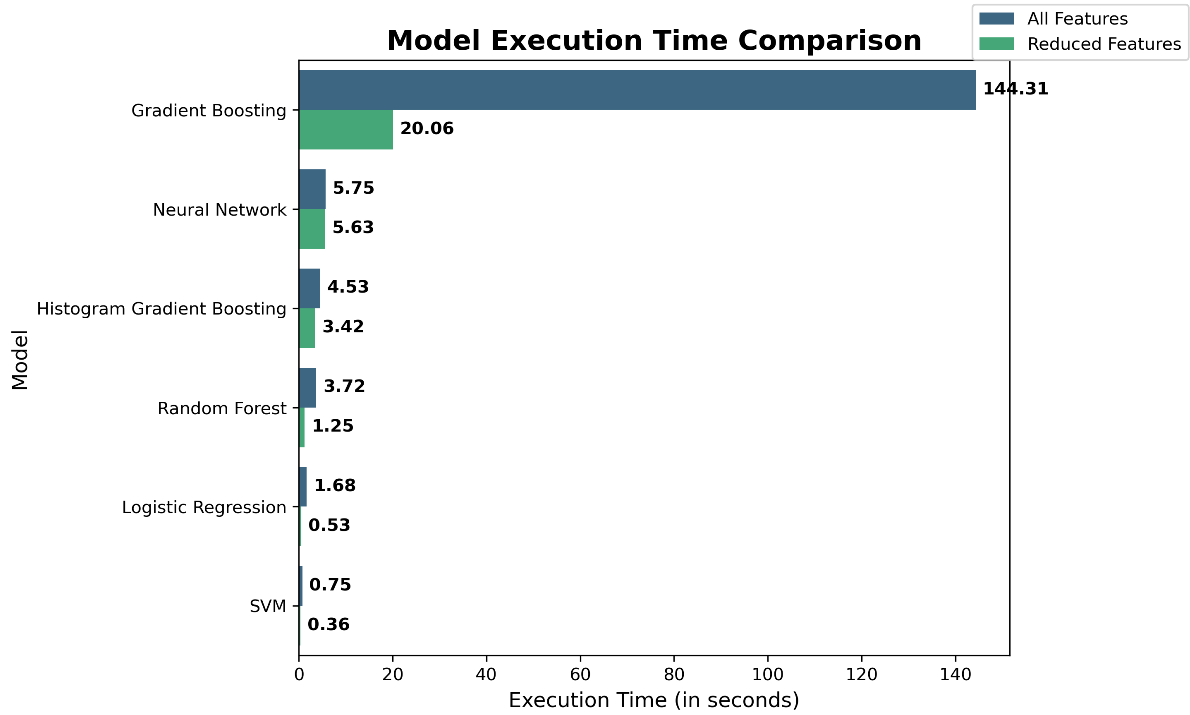

Figure 9: Execution Time

Looking at the execution times, the value of the feature reduction is very apparent. All models experienced a reduction in time to execute, albeit to different extents. The standard gradient boosting model experienced the greatest reduction (86%), which is logical given its overall slow performance and dimensionality sensitivity. Other models, like the random forest, logistic regression, and SVM all experienced great reductions as well, and had very low execution times.

Figure 10: F1 Scores

The F1 scores show us that the the slight reduction in accuracy comes mostly from the walking activities, as the other categories are pretty consistent from before. For logistic regression, the score falls near the levels of sitting and standing, showing a general struggle to separate between the activities. For the rest, the reduction in score is very minor, and they maintained > 0.96 in those categories. Overall, this shows that the feature reduction was highly successful in reducing the features while maintaining accuracy and lowering execution time.

5 Conclusion

In conclusion, all of the machine learning models we tested were able to classify activity among the 6 categories with high accuracy. This is especially true when considering static and dynamic activities, but is more difficult for activities with similar signals.

Our analysis shows that Histogram Gradient Boosting is the optimal model for this task, offering the best balance of training speed and classification accuracy on difficult classes. In general, the ensemble methods outperformed the traditional models, and show their ability to identify more nuanced differences between the classes. The linear nature of the logistic regression and support vector models did inhibit their performance in the end, particularly after feature selection. However, they still showed very high performance at very low execution times. With more data or different classes, these models could be very useful when execution time is important.

Future work on this topic could involve a more in-depth look at hyperparameter tuning, particularly for the MLP model. Using a different model structure or using

Recurrent Neural Networks (RNNs) to exploit the time-series nature of the data could prove to be interesting. Further, adding more classes (running, swimming, etc.) could also be interesting and further test the capabilities of these models.

6 References

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